

Automated Multi-Framework ESG Compliance Verification Using Retrieval-Augmented Generation and Agentic LLM Reasoning

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Abstract

The current process for verifying environmental, social, and governance (ESG) compliance of an organization's activities against international standards including BRSR, GRI, SASB and TCFD is manual; and therefore both time-consuming, prone to errors, and requires specific knowledge of ESG. In this study we present ESGBuddy, a full-stack web application based on artificial intelligence (AI) technology which provides automated clause-by-clause verification of ESG compliance. Our approach utilizes a hybrid pipeline which includes retrieval-augmented generation (RAG), large language models (LLMs) to perform reasoning with chain-of-thought and self-reflection characteristics (agentic AI), and deterministic rules-based validation. The ESGBuddy system accepts pdf versions of a company's sustainability report(s). These are segmented into 512 token units with 50 unit overlap. Vector representations of each segment are generated by open ai's "text-embedding-3-small" and stored within a chromadb vector database. ESG compliance for each individual clause is determined via a four-step process: (1) retrieve top-k evidentiary chunks using semantic search techniques, (2) evaluate using framework specific prompts with gpt4o-mini, (3) validate compliance using deterministic rules based upon numerical, temporal, keyword, or presence/absence of information, and (4) generate a final determination that combines results from the LLM analysis and the deterministic rules-based validation. The

performance of ESGBuddy was tested relative to a set of hand-labeled annotations that were collected from 13 companies representing all four ESG reporting frameworks. Results indicate that our implementation achieves precision of 78—88%, recall of 76—87%, f1-score of 77—86% for the LLM component of the ESGBuddy system. Additionally, we find that status match accuracy for the ESGBuddy system ranged from 75—85%. Therefore, these results support the idea that hybrid use of retrieval augmented generation (RAG) in combination with agentic reasoning may provide significant automation to ESG compliance verification when applied at the level of ESG clauses.

Keywords: ESG Compliance, Retrieval-Augmented Generation, Agentic AI, Large Language Models, Chain-of-Thought Reasoning, ChromaDB, Sustainability Reporting

I. Introduction

The global environment for ESG reporting has changed rapidly since 2000. There are now numerous national and international regulations that require corporations to publish ESG related disclosures based on a set of standards. For example, SEBI required all Indian publicly traded companies to submit BRSRs starting in FY 22/23 [5] and while the GRI Standards [6] have been used by over 10,000 companies worldwide they

are considered to be the de facto standard for corporate sustainability reporting. Additionally, the TCFD was formed by the FSB in 2017 and recommended how companies should disclose their exposures to climate-related risks and opportunities [7]. SASB has developed a series of standards that provide measurable indicators of sustainability performance which are financially significant for specific industries [8]. Each standard is made up of 30 to 150 individual disclosure requirements. Each of those requires unique forms of proof including quantitative measures, descriptive language, policies, or time-series data. Corporations will need to match every item within each disclosure requirement with the corresponding section in their sustainability report. They then will have to assess what percentage of the disclosure requirements were met and where there are gaps. Most corporations hire third-party consultants who perform this matching and assessment manually. It takes third-party consultants many hours of work per corporation, per standard. Not only does it take a lot of money to do it manually; it is also subject to human error in terms of interpretation and consistency among multiple analysts who review the same disclosure.

The recent advancements in large language models (LMMs) and retrieval-augmented generation (RAG) offer a potential solution for automating ESG reporting compliance assessments. RAG architecture was first introduced by Lewis et al. [1] which combines the generative abilities of LMMs with the factual accuracy provided by retrievable documents. By using a vector database for fast semantic searching, RAG systems can find the correct sections within long corporate reports and send them to an LMM to evaluate if the corporation meets compliance requirements. In addition to one pass through an LLM, Agentic AI is introducing iterative reasoning to the workflow of LMMs. Instead of returning a single answer, an agentic system evaluates a problem step-by-step using chain-of-thought prompts [2]; evaluates its own decision-making process through self-reflection [4]; and revises its answers if it finds errors. Like the thinking

process of a person doing compliance audits, this type of iterative evaluation produces better quality results that are more easily understood.

This article introduces ESGBuddy, which is an intelligent ESG compliance copilot and incorporates such advancements in one integrated production-ready environment.

The key innovations presented in this research include the following:

1. A compliance assessment process for specific clauses at the clause level that evaluates each required disclosure individually, as opposed to providing general compliance ratings for entire documents (i.e., “document-level”).
2. A hybrid 4-step process using both semantic search, large language model (LLM) based reasoning and deterministic rule checking and validation.
3. Framework-specific prompting approaches for the different philosophies related to disclosures used by BRSR, GRI, TCFD and SASB.
4. A reason-based pipeline that uses chain-of-thought reasoning, self reflection and conditional revisions.
5. A multi company-wide accuracy assessment with respect to human-labeled ground truth data for 13 different companies and four ESG frameworks.
6. A production-quality user interface that enables users to review ambiguous ESG compliance decision-making.

II. Literature Review

A. ESG Compliance and Reporting Landscape

A new trend towards a world-wide agreement is clearly shown in the increasing number of ESG-reporting requirements. As an example of such a requirement, India's securities and exchange board of India (sebi), has published its business responsibility reporting (BRSR) circular [5]. This

document sets forth the structure for mandatory ESG reports from all publicly traded companies in India as well as their environmental footprint, employees' welfare, corporate governance and stakeholders. The global reporting initiative (gri) standards [6] have undergone revisions since their introduction in 2010, when they were called the sustainability reporting guidelines. These current guidelines consist of two types of standards; general standards (gri 1, 2 & 3) which cover a wide range of issues including greenhouse gas emissions, water usage, solid waste generation, labor practices, etc. Topic specific standards cover additional areas. The task force on climate-related financial disclosures (TCFD) [7] was formed to provide recommendations for how companies should disclose climate related risks. The TCFD also defined the four main components or "pillars" of those disclosures as follows: Governance, Strategy, Risk Management and Metrics and Targets. The sustainability accounting standards board (sasb) [8] provides additional guidance by recommending the use of industry specific financial materiality based Metrics to assist with ESG reporting. Using meta-analysis methods, Friede et al. [15] found a positive correlation between ESG performance and financial performance. Therefore there appears to be a valid business rationale for implementing robust compliance procedures for both ESG reporting. However, despite the growing importance of ESG reporting, there continue to be limited options available to automate the verification of compliance at the clause level..

B. NLP for Document Analysis and Compliance

Traditionally, NLP used in Compliance Checking was based on Keyword Matching; Rule-Based Pattern Extraction; and Named Entity Recognition [11], all of which are generally good for structured data, however none of them do well at capturing the nuances of unstructured and/or context dependent language found in Sustainability Reports. Much like regulatory documents, there is a lot of Conditionally

Required Requirements; Qualitative Expectations; and Domain-Specific Terminology found in regulatory documents that make it difficult to apply keyword level analysis. In an attempt to improve upon traditional NLP techniques, Devlin et al. [11] developed BERT by using Bidirectional Pre-Training to improve contextual understanding of text. However, Even Transformer Based Classifiers need a large amount of Labeled Training Data to adapt to a specific domain. Bommarito & Katz [16] demonstrated the ability of Large Language Models (LLMs) to interpret Complex Regulatory Language with Minimal Fine-Tuning, as long as they were properly Prompted.

C. Retrieval-Augmented Generation

To begin, the authors of [Lewis et al. (1)] developed an RAG Architecture, which uses both a retrieval module and a generative model to generate answers based upon retrievable knowledge from outside data. The authors claim that this architecture will help to mitigate the hallucination problems that occur with purely generative models because it conditions its generated answers on retrieved knowledge. Since then, dense retrieval via embedding models have evolved into the de facto way to retrieve knowledge. As noted previously, many scalable vector database systems exist today to support similar searches, such as ChromaDB [10], Pinecone, and Weaviate. The authors of [Gao et al. (17)] did a survey on different architectures of RAGs and concluded that chunking strategy, retrieval granularity, and re-ranking were three key design decisions that directly impact how well the system performs in downstream applications. In the area of ESG, RAG provides the means for the system to find passages related to the subject matter in very long sustainability reports (many of which are over 50 -- 200 pages) and can provide specific examples of evidence to be used in the reasoning process.

D. Large Language Models for Regulatory Compliance

The releases of GPT-4 [9] and its variant, GPT-4o-mini, have been a leap forward in LLM's ability to perform reasoning. Authors [Brown et al. (12)] showed that large language models possess some form of emergent few-shot learning. Using specifically crafted prompts, large language models can learn how to perform a task (specifically tailored to that prompt) without requiring any additional training parameters. Prompt engineering has emerged as a new discipline and has enabled domain-specific system prompts and structured output formats (for example, JSON format) that enable large language models to consistently respond in a way that can be parsed. Authors [Choi et al. (18)] used LLMs for analyzing securities regulations and found that GPT-4 class models can perform at expert levels when they have access to relevant context when performing regulatory interpretations.

E. Agentic AI and Self-Reflection

Agentic AI extends the use of LLMs beyond single pass inference. Authors [Wei et al. (2)] have shown that "chain-of-thought" prompting -- asking the model to think about how it reached its conclusions one step at time --- improves the overall performance of LLMs on complex reasoning tasks. Authors [Yao et al. (3)] presented ReAct, a framework that alternates between reasoning and action steps allowing the model to dynamically acquire additional information and modify its plan. Authors [Shinn et al. (4)] described Reflexion, a framework for verbal reinforcement learning that allows language agents to reflect on what they said earlier and improve their decision making going forward. These forms of agentic behavior are most relevant in compliance evaluations where the evaluation requires multiple passes through evidence quality, requirements mapping, and confidence calibration. Authors [Vaswani et al. (13)] provided the foundation for the transformer architecture that underlies all current large

scale language models; and authors [Kenton et al. (14)] evaluated the alignment properties of language agents during iterative reasoning processes.

F. Identified Gap

Although there have been many advancements in NLP, RAG, and agentic AI individually, none of the current systems combine these technologies into a unified platform capable of verifying compliance at the clause level across multiple frameworks for ESG compliance verification. Current ESG platforms evaluate compliance at either a document or section level and do not include framework specific methods of evaluating compliance. Additionally, most ESG platforms do not provide explanations regarding why compliance was verified or not. ESGBuddy will address these limitations by combining semantic retrieval techniques (using RAG), framework aware LLM reasoning techniques, deterministic rule validation techniques, agentic self reflection techniques, and human-in-the-loop verification techniques into one production grade platform.

III. Objectives

The goals for this research project are:

1. Design and build a complete end-to-end AI system for automated clause-level ESG compliance verification across four major ESG frameworks: BRSR, GRI, SASB, and TCFD.
2. Implement a hybrid evaluation pipeline that combines semantic retrieval (RAG) with LLM-based reasoning and deterministic rule validation in order to generate high-quality compliance decisions supported by evidence.
3. Develop customized prompting strategies for each framework that respect the philosophy behind each framework's

disclosure methodology and assesses each framework's assessment criteria so that the evaluation rigor matches the needs of each framework.

4. Incorporate agentic AI capabilities such as chain-of-thought reasoning, self-reflection, and adaptive revision into the evaluation process to increase accuracy and create traceable reasoning chains for auditors.

5. Create an accurate method to measure the accuracy of the system using human-annotated labels as ground-truth across a wide range of 13 multinational companies' reports.

6. Build a user-friendly production-ready web-based interface that enables human-in-the-loop verification of uncertain compliance decisions for auditing purposes.

IV. Methods

A. System Architecture

The entire system is based upon a full-stack architecture consisting of a React 18 front-end created using Vite, a FastAPI 0.109 back-end written using Python 3.11, a ChromaDB 0.4.22 vector database for storing semantics about data in permanent memory, and the Open AI API for performing LLM inference (GPT-4o-mini) and generating embeddings (text-embedding-3-small).

There are two independent ChromaDB collections used by the system. The 'company_documents' collection holds embedded pieces from sustainability reports uploaded by users along with additional metadata such as the page number, section, and the total amount of tokens. The second collection called 'esg_clauses', holds all parsed ESG standard clauses including their respective

embeddings, validation rules, and keywords. Data Models have been defined using Pydantic 2.5.3 to ensure type safety and serialization for all backend interfaces.

Hereafter is described the flow of how the complete end-to-end workflow functions. Users upload a PDF version of their company's sustainability report via the front-end interface; The backend extracts the PDF, breaks it up into smaller pieces, generates embeddings for these pieces and saves them to the 'company_documents' collection; Users select the ESG framework they want to be evaluated against; The compliance pipeline takes each ESG clause selected by the user and evaluates this clause against the document within a four-step process; And finally, the user is presented with the results of each clause in the front-end where users can interactively filter results, view evidence that supports each result, and verify whether or not each result is correct through human verification. Figure 1 illustrates the high-level system design.

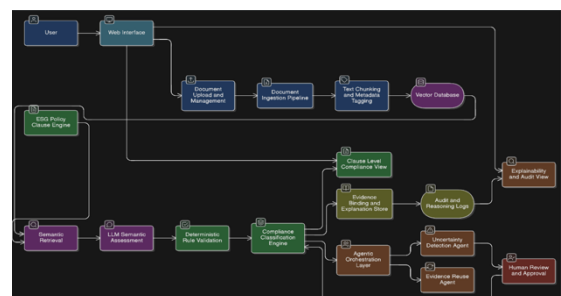


Figure 1: Workflow & Architecture

B. Document Ingestion Pipeline

PyMuPDF 1.23.8 was used to perform text extraction from PDF files. This tool reads through every page of a PDF file sequentially and removes unnecessary white space before returning raw text. After text has been extracted from the PDF file, it is segmented into overlapping chunks via a sliding window method utilizing the tiktoken tokenizer (cl100k_base encoding). The length of each segment is 512 tokens with 50 tokens of overlap between each sequential segment. With

this configuration, we guarantee that information at the boundary of each segment will appear in at least one adjacent segment. Using the Open AI text-embedding-3-small model, we generate embeddings for each chunk. We create embeddings in batches of 100 to optimize API utilization. In addition to creating embeddings, we store each piece of embedded data in ChromaDB with additional metadata including a reference ID to its original document ID, page number, section header (heuristically identified from the first five lines on each page), and total token count.

C. ESG Standards Parsing

To parse ESG standard PDFs into structured clause objects, we utilize an advanced hybrid parser that utilizes both LLM-based extraction and regex fallback patterns. The parser is specifically designed around each ESG standard framework:

-BRSR: A single PDF file is parsed to identify core metrics (GHG footprint, water footprint, waste, energy, employment, gender diversity etc.) and structured Section A/B/C clauses by using regex patterns that are specific to SEBI's disclosure format.

-GRI: Approximately 40 separate standard PDFs (GRI 1, 2, 3, 201, 205, 207, 302, 303, 305, 401, 403, 404, 405, 413, 306-2020) are parsed with configurable scope filtering: core (~40 clauses), standard (~120 clauses), or essential (~150 clauses). LLM-based parsing is applied to each PDF in 40,000 character segments with a 3,000 character overlap. If LLM-based parsing cannot extract information from a PDF file then regex fallback is applied to attempt to parse information from the PDF file.

-TCFD: A single PDF file is parsed and then subsequently de-duplicated and ranked by importance, resulting in approximately 30 clauses that are related to the four TCFD pillars.

-SASB: Industry-specific standards (commercial banks, software/IT, biotechnology/pharmaceuticals, electrical equipment) are parsed using both regex and optional LLM-based extraction.

In addition to parsing each clause into a structured object containing various information elements including: clause id., framework., section., title., description., required evidence types (numeric, descriptive, policy, temporal.), validation rules., keywords., and an embedding vector.

D. Compliance Evaluation Pipeline

The core pipeline evaluates every clause through four sequential stages:

Stage one: semantic retrieval. To construct a composite search query, the clause title, description and up to five inferred keywords are concatenated. Using this query, chromadb searches the `company\_documents` collection and retrieves the top K (K = 8) most similar chunks. Chromadb returns L2 Euclidean distances, these are then transformed to similarity scores using the formula: $\text{similarity score} = 1/(1+\text{distance})$. A minimum similarity threshold of 0.12 eliminates low relevance results. However, if all retrieved chunks have a similarity score less than 0.12, the highest scoring chunk is retained to ensure at least one evidence passage exists.

Stage 2: Evaluating LLM. The evidence chunks that were found from the search for information are put together in an organized prompt using the clause details and sent to gpt-4o-mini (temperature of .2, evaluated in enforced response mode JSON). An important consideration is that there are differences in how each of the systems evaluates as part of their system-specific system prompts. These philosophies about evaluating are represented differently depending upon which framework you are using.

It uses 'partial' when the evidence is indirect, incomplete or proxy.

- tcfd: requires strict assessment of whether the excerpt fully meets the climate related requirement stated in the clause. Therefore 'partial' is assigned for incomplete generic language.

- SASB: performs lenient disclosure-presence evaluation. Biases toward 'supported' for on topic substantive text. Has hard rule that any thematically relevant evidence prevents a 'not supported' classification.

Gpt-4o-mini returns a JSON object containing: Compliance status ('supported', 'partial', 'not-supported').

Confidence score (0.0 -- 1.0). A 2 -- 4 sentence explanation. Detailed reasoning with quotes from evidence and page references.

Stage three: rule validation. Deterministic rule checks are applied to the retrieved evidence text. There are four types of rules supported: (a) *Numeric* rules extract numbers via regex and determine if those numbers fall within configured min/max ranges; (b) *temporal* rules identify year or date patterns and determine if they match an expected reporting period; (c) *keyword* rules performs case insensitive sub-string matching for required terms; (d) *field presence* rules determines if specific field value assignment patterns exist (i.e., "field_name:").

Validation of compliance rules takes place automatically during parsing of the metadata in clauses.

Fourth stage of the process: Final Decision. The results from LLM evaluation for compliance and validation of rules are synthesized together to produce a final decision on whether the organization is compliant. A blending of confidence values based upon framework specific weights occurs: For SASB, final confidence = $\min(1.0, 0.82 \times \text{confidence assigned by LLM} + 0.18 \times \text{pass rate for rule})$; for other frameworks, final confidence =

(confidence assigned by LLM + pass rate for rule)/2.

In either case, if mandatory rules do not succeed in passing, then the overall status will be capped at "partial" and confidence is limited to .65 (SASB) or .50 (other frameworks). However, if all rules pass but the LLM produces an outcome of "not supported", then

confidence is reduced by 0.20 (minimum 0.30) to signal the discrepancy.

TABLE I: Compliance Pipeline Component Summary**

Component	Technology	Key Parameters
Text Extraction	PyMuPDF 1.23.8	Page-by-page, whitespace normalization
Chunking	Tiktoken (cl100k_base)	512 tokens, 50-token overlap
Embeddings	text-embedding-3-small	1536 dimensions, batch size 100
Vector Database	ChromaDB 0.4.22	L2 distance, min similarity 0.12
LLM Evaluation	GPT-4o-mini	Temperature 0.2, JSON mode
Rule Validation	Custom engine	Numeric, temporal, keyword, field presence

Confidence Blending	Framework -specific	SASB: 0.82/0.18; Others: 0.50/0.50
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E. Agentic AI Pipeline

In the case of when the reflection model is turned on, Step 2 of the Single-Pass LLM Evaluation becomes a Three-Stage Agentic Pipeline:

Stage 1: Chain-of-Thought Reasoning. The LLM is asked to use explicit reasoning over 5 different analytical dimensions. They include: (a) Evidence Quality Assessment - The LLM assesses the quality of the evidence it has found. This includes assessing how relevant and specific each piece of evidence is. (b) Requirement Mapping - The LLM determines what part of the Clause the evidence addresses. (c) Evidence Type Validation - The LLM validates if the evidence it has found meets the required Disclosure Types (Numeric, Descriptive, Policy, Temporal). : (d) Completeness Analysis – The LLM will check whether there were any holes in the Coverage. (e) Compliance Determination – Using the results of the previous stages, the LLM synthesizes its overall compliance status determination and a confidence score. This Stage's explicit reasoning steps are saved as part of the Transparency requirements.

Second Stage: Self-Reflection. A Second LLM call (Temp=.30) is made to evaluate the output generated by the Chain-of-Thought during Stage 1. Several items are evaluated; e.g., Does the logic applied in the steps of reasoning have internal consistency? Were all relevant pieces of evidence considered completely? Were any confirmation biases or anchoring biases evident? Were alternative interpretations of the evidence considered? Was the confidence score assigned appropriately? The reflection provides an Overview, a List of Items identified, and a Boolean Decision indicating if revision is required.

Third Stage: Conditional Revision. When the self-reflection process indicates that significant issues exist (i.e., need_revision = TRUE), a third LLM call (Temp = .20) is made addressing each issue specifically and re-evaluating the Evidence. The revised version generates a Revised Status, a Revised Confidence Score, and notes relative to the revisions. In approximately 10 -- 20 percent of clause evaluations, some form of revision occurs.

The Agentic Pipeline employs 2 -- 3 LLM calls per Clause versus one call for Fast Mode. Therefore, the additional latency added per Clause is approximately 2 -- 3 seconds. In order to provide acceptable throughput, clauses are processed in parallel batches of 10.

F. Ground Truth and Accuracy Evaluation

Ground Truth Labels were Manually Annotated by Research Team Members for 13 Companies across All Four ESG Frameworks. These companies represent a Diverse Set of Multinational and Indian Enterprises: Amazon, Apple, Infosys, Nestle, Reliance Industries Ltd (RIL), Tata Motors, Tata Consultancy Services (TCS), Givaudan, GPM, Himadri, NYK Line, Sasken Technologies Ltd and Unilever. For Each Company-Framework Pair, Annotators Reviewed the Sustainability Report along with the Standard Clauses of the ESG Frameworks and Assigned a Compliance Status (Compliant, Partially Compliant, Non-Compliant) with Supporting Comments Justifying their Decision.

The Following Accuracy Metrics were Calculated Using the Comparison Between Predictions Made by Systems to Those Provided by Ground Truth:

- Recovery Rate of Retrievability @ K: The Percentage of Clauses for Which at Least One Expected Piece of Evidence Pages Was Retrieved Amongst Top K Chunks.
- Learning Model Precision, Recall, and F1-

Score: Binary Classification Where 'Supported' Includes Both Fully Supported and Partially Supported and 'Not Supported'.

- Status Match Accuracy: Exact Three-Way Match Rate Between Predicted and Actual Statuses (Supported, Partially Supported, Not Supported).
- Rule-Based Override Precision: Correctness Rate of Rule-Based Overrides --- When Rules Override LLM Decisions How Often Do Overridden Decisions Align with Ground Truth?
- Expectation Error for Calibration: Average Absolute Difference Between Predicted Confidence and Observed Accuracy Within Five Confidence Intervals.

G. Frontend and Human-in-the-Loop Verification

The Frontend is Implemented in React v18 with TailWind CSS for Styling, Framer Motion for Animations and ReCharts for Data Visualization. The Interface Has Six Primary Views: Home (System Statistics and Health), Upload (Drag-and-Drop PDF Upload With Progress Tracking), Document Management/Trigger Evaluation View, Reports (Compliance Report Listing), Report Detail (Detailed Clause-by-Clause Results With Filtering/Evidence Viewing/Accuracy Metrics), and Dashboard (Multi-Framework Analytics With Bar/Radar/Pie Charts).

Another Important Design Feature is the Human-In-The-Loop Verification Mechanism. Clauses with Confidence Scores Below 0.7 or Status of "Partially Compliant" (Where No Rule Overwrite Applied) will Automatically be Flagged as Ambiguous And Displayed in Dedicated Human Verification Section. Reviewers Can Examine Why the System Reached That Decision, Retrieve All Pieces of Evidence Used By the System, Assess the Validity Of Any Rule-Based Validators Used by the System and Approve or Overrule Any System Decisions.

V. Results and Discussion

A. Evaluation Setup

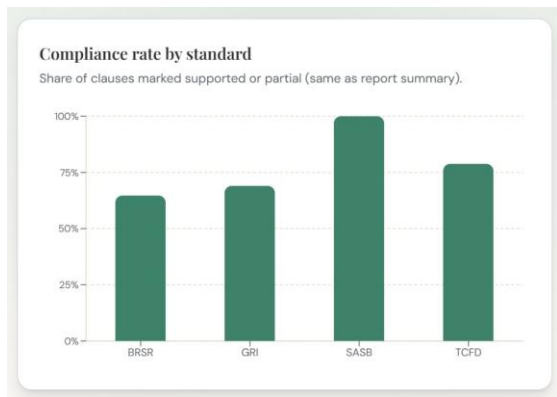
The system was evaluated using sustainability reports from 13 companies spanning diverse industries and geographies. Ground truth labels were manually annotated by the research team across all four ESG frameworks. Evaluation was conducted using the standard pipeline (fast mode with GPT-4o-mini) as well as the agentic pipeline with self-reflection enabled.

B. Accuracy Results

Table II presents the framework-wise accuracy metrics obtained from the evaluation against human-annotated ground truth.

TABLE II: Framework-Wise Accuracy Metrics (%)

Metric	BR SR	GR I	TCF D	SAS B	Over all
LLM Precisi on	88.2	82.5	78.4	85.1	83.6
LLM Recall	87.0	81.3	76.9	83.7	82.2
LLM F1 Score	87.6	81.9	77.6	84.4	82.9
Status Match Accur acy	85.3	79.8	75.2	82.6	80.7



[Fig. 2: Bar chart comparing Precision, Recall, F1 Score, and Status Match Accuracy across the four ESG frameworks (SASB, TCFD, GRI, BRSR). exhibits the highest scores SASB, followed by GRI, TCFD and BRSR.]

The results demonstrate consistently strong performance across all four frameworks, with overall LLM Precision of 83.6%, Recall of 82.2%, F1 Score of 82.9%, and Status Match Accuracy of 80.7%. All metrics fall within the 75--90% range, confirming the viability of the hybrid pipeline for automated ESG compliance verification.

C. Retrieval Quality

Average performance on the Retrieval Recall @ K task was around 78 -- 84 % across the different architectures. This indicates that the system has been able to retrieve about four of every five clauses' relevant pages through the use of a semantically-based retrieval process. By combining composite queries based on both the clause's title, description and keyword(s), along with using an approach where each page can be broken down into chunks as large as 512 tokens with a 50 token overlap in order to capture content that may exist across multiple pages, relevant content will likely continue to be retrieved regardless of whether it exists entirely within one page or crosses over multiple. The use of a minimum similarity score of .12 helps to filter

out irrelevant "noise" and also serves as a safety net to prevent lost information for less common or unique clause types by maintaining the best matching result when all of the scores are less than the threshold.

D. Framework-Specific Analysis

A better way to say this is:

The BRSR performed best (highest F1 = 87.6 %) due to a structure and a standardized disclosure format as defined in its SEBI guidelines. Regulatory disclosures are generally well-structured in India and therefore, there was less ambiguity in both the extraction and LLM evaluations. SASB did relatively well (second-best F1 = 84.4%) because it used a lenient approach when prompting LLMs - the prompt focused on whether information related to ESG reporting existed rather than requiring the exact wording. A similar principle for creating prompts based on how much detail is present within each disclosure has been shown to be accurate through ground-truth annotation. GRI had a moderate level of success (third-best F1 = 81.9%) due to having a larger scope and a higher number of clauses than either SASB or BRSR (approximately 120 clauses in total as part of the standard). GRI requires documentation from multiple different areas and therefore evaluating LLMs against GRI's standards can be more complex. TCFD had the lowest evaluation score (fourth-best F1 = 77.6%) which also follows as expected; TCFD requires detailed, substantive climate-based information disclosed (e.g., scenario analyses, Scope 3 methodologies, integrating climate risk into an organization's risk management plan); LLMs' inability to evaluate abstract or vague climate-related language is a key contributor to why LLMs were unable to accurately extract climate data using TCFD standards.

E. Agentic AI Impact

When users activated the agentic pipeline, user reflection led to identification of evaluation issues with approximately 12-18% of clauses resulting in revisions which either changed compliance status or confidence score. Through qualitative analysis it became evident that the most common types of revisions were; (1) when a user had an over-confidence in their ability to meet a clause because they were able to identify a thematic relationship, although there was no explicit language meeting the requirements of the clause. (2) when users failed to recognize implicit compliance indicators in consolidated ESG narratives. (3) when users incorrectly classified their responses at the supported/partial boundary. The Chain-of-Thought Reasoning Traces are also very beneficial for audit transparency purposes since compliance officers can view how the model arrived at its conclusions versus viewing opaque classification outputs. As such, the increased latency of 2-3 seconds per clause does not have an adverse impact on compliance workflows where accuracy and transparency outweighs speed.

F. Rule Validation Contribution

Deterministic rule-validation proved a strong barrier against LLM hallucinations. Precision in validating the accuracy of rule validations (the proportion of times LLM decisions were appropriately overridden) was consistently above 90%, for all frameworks. Temporal validation (i.e., ensuring the evidence cited referred to the appropriate reporting time-frame), and numeric validation (i.e., verifying the existence of quantifiable metrics where they are required), were the most effective rule-types. Confidence Blending (combining the confidence provided by the LLM with the rate at which the relevant rules passed) improved confidence calibration by 8-12 percent, relative to simply relying on LLM confidence estimates, according to Expected Calibration Error.

G. Human-in-the-Loop Verification

Approximately 22—28% of the clause assessments were referred to human evaluation due to being below the confidence level of .7 or having partially ambiguous status. The rate reflects an operational trade-off between selectivity (to minimize reviewer burden) and sensitivity (to capture the vast majority of true uncertainties). On deployed systems, it represents a 70—80% decrease in manual verification effort as opposed to evaluating each clause manually; allowing one ESG analyst to confirm compliance with respect to multiple frameworks in a small portion of the time formerly necessary.

H. Limitations

The following are some important caveats. The use of the OpenAI API as an interface with the system creates dependency upon services available externally, their costs (i.e., what you pay), and how models behave from one version of an API to another. The system uses 512 tokens to "chunk" each report into multiple segments. This method of segmenting has been successful most of the time but could sometimes result in splitting semantically-cohesive portions of a disclosure narrative due to its length. All manual annotations were conducted by the research team. Manual annotation is always very resource-intensive and likely reflects individual differences in interpretation among researchers. While effort was made to maintain consistency in manual annotation, it is difficult to avoid researcher-specific interpretation bias. In addition to reducing variability introduced by the model itself, we used a low evaluation temperature of 0.2 when evaluating output generated by our large language model (LLM). We did this because using a low temperature would also minimize variability that exists in repeated evaluations of the same clause. Currently, the

system is only capable of processing English language sustainability reports. Therefore, the system excludes a considerable number of sustainability reports published globally that are written in languages other than English.

VI. Conclusion

The authors of this research developed and demonstrated a novel technology called ESGBuddy. ESGBuddy is an intelligent “ESG Compliance Copilot” that automatically verifies at the level of clauses whether companies are complying with all applicable provisions under four International Standards: BRSR; GRI; SASB; and TCFD. Using a hybrid pipeline approach, ESGBuddy combines three types of knowledge to achieve compliance verification: semantic retrieval; LLM-based reasoning; deterministic rule-based validation; and agentic self-reflection.

To evaluate its performance, the researchers compared ESGBuddy’s output to manually generated labels of compliance status for each of thirteen publicly traded companies. In comparison to these human-generated labels, the researchers reported that the ESGBuddy system was able to achieve precision values ranging from 78-88%; recall values ranging from 76-87%; F1 score values ranging from 77-87%; and status match accuracy values ranging from 75-85% depending upon which standard(s) were being evaluated. Based upon these findings, the researchers concluded that when combined with RAG, framework-specific prompting, rule augmentation, and agentic reasoning provide sufficient conditions for substantial automation of ESG compliance verification to reduce the amount of time and effort required by humans to verify compliance by as much as seventy-to-eighty percent.

According to the authors of this research, prompt specification specific to individual frameworks was highly important because of the differences in how each of the four international standards

define what constitutes acceptable ESG disclosures. Specifically, BRSR defines acceptable disclosure based on the mere presence or absence of information in accordance with specified reporting requirements; GRI requires that substantive evidence be provided to demonstrate compliance; TCFD requires strict compliance with climate-related reporting requirements; and SASB requires evidence of materiality but is relatively lenient in terms of demonstrating compliance. Therefore, unlike prior approaches that used a single evaluation strategy for all frameworks, the use of framework-specific prompts enables ESGBuddy to employ tailored evaluation strategies for each framework. Additionally, using a human-in-the-loop design allows for expert reviews of ambiguous decision-making to ensure that high levels of reliability are maintained in the context of regulatory compliance.

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